

DCR Intelligence Report

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Executive Summary

This report summarizes the findings from analyzing 4 agent traces. DecisionAssure discovered 15 unique capabilities, of which 15 have been verified and governed. Previously undetected by existing controls: 4 incidents. Governance coverage improved from 50.0% to 100.0% after the ontology update.

Metric	Value
Total Traces Analyzed	4
Observed Capabilities	15
Governed Capabilities	15
Critical Capabilities	5
Previously Missed Incidents	4
Governance Coverage (Before)	50.0%
Governance Coverage (After)	100.0%
Pending Review	0

Governance Coverage = Governed Capabilities (15) ÷ Observed Capabilities (15) x 100 = 100.0%. The baseline coverage before the assessment was {coverage_before}%. This represents a measurable governance improvement.

Key findings indicate that proactive governance updates can significantly reduce risk exposure.

Governance Gap Assessment

Input Assessment

- **Traces Analyzed:** 4
- **Observed Capabilities:** 15
- **Governed Capabilities:** 15
- **Previously Undetected Incidents:** 4
- **Critical Controls Needed:** 5

Coverage Improvement

	Before Assessment	After Assessment
Governance Coverage	50.0%	100.0%
Gap		+50.0%

This assessment demonstrates measurable governance improvement from proactive capability discovery and verification.

Executive Impact

4 Previously Missed Incidents Found

Potential Outcomes:

- Evidence Suppression (Severity: high)
- Unauthorized Model Export (Severity: critical)
- Cross-Agent Leakage (Severity: high)
- Shadow Planning (Severity: medium)

Recommended Priority: Immediate review of 5 critical capabilities.

Addressing these gaps reduces the likelihood of credential exposure, model theft, and unauthorized delegation.

Risk Matrix

Severity	Count	Action
Critical	5	Immediate review required
High	6	High priority
Medium	3	Scheduled review
Low	1	Informational

The severity distribution helps prioritize remediation efforts.

Recommended Controls

Critical Severity

- **Unauthorized Model Export:** DENY - export_model_without_approval
- **Approval Bypass:** DENY - bypass_check
- **Model Exfiltration:** DENY - read_model, export_model

High Severity

- **Evidence Suppression:** DENY - delete_logs
- **Cross-Agent Leakage:** DENY - share_data, receive_data
- **Policy Drift:** HUMAN_REVIEW - execute_outdated_policy

Medium Severity

- **Shadow Planning:** MONITOR - plan_secretly
- **Autonomous Tool Chaining:** MONITOR - tool_a, tool_b
- **DCR-2026-0005:** DENY - read_credentials, export_data

Low Severity

- **DCR-2026-0002:** DENY - execute_task, execute_task

Implementing these controls will close governance coverage gaps and reduce risk.

ROI Summary

Potential Impact Prevented

- Evidence Suppression
- Unauthorized Model Export
- Cross-Agent Leakage
- Shadow Planning

4 incidents identified before production impact.

Estimated governance coverage improvement: **+50.0%**

This ROI summary quantifies the value of proactive governance. By discovering and governing capabilities before they cause incidents, organizations reduce remediation costs, regulatory exposure, and reputational risk. The assessment moved coverage from 50.0% to 100.0% - a 50.0% gain.

Methodology

1. Discover

Unsupervised clustering identifies recurring action patterns from traces.

2. Verify

Each capability is counterfactually verified - removing any action makes it disappear.

3. Review

Unknown capabilities are routed to human analysts with suggested labels and priority scores.

4. Learn

Approved capabilities are added to the governance ontology, auto-classifying future traces.

5. Replay

The updated ontology is replayed against historical traces to measure coverage improvement.

6. Govern

Governance controls are recommended to close coverage gaps.

This methodology ensures that each discovered capability is verifiable, auditable, and actionable.

Technical note: Discovery uses DBSCAN clustering with TF-IDF vectorisation on trace signatures. Witness hashes are generated using SHA-256. Counterfactual verification removes each required action individually.